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B1601F81734886B96D74577AE4DC03B2

SWIFTROUTE

Project Overview

The *SwiftRoute Logistics* project focuses on analyzing delivery performance and fleet operations using IBM Cognos Analytics. The main goal was to help the Fleet Manager monitor key operational challenges such as:

- Rising fuel costs
- Delivery delays and failures
- Vehicle breakdowns
- Shipment damages and customer dissatisfaction

By transforming raw logistics data into meaningful insights, the project enables data-driven decision-making across operations.

1. Introduction

SwiftRoute Logistics operates a fleet of 500 vehicles across 20 states in India, serving industries such as e-commerce, FMCG, and pharmaceuticals. The company faces operational challenges including increasing fuel costs, delivery failures, vehicle breakdowns, and customer complaints.

This project uses IBM Cognos Analytics to analyze logistics data and provide actionable insights for improving operational efficiency and decision-making.

2. Objectives

The primary objectives of this project are:

- To monitor delivery performance and SLA compliance
 - To analyze vehicle breakdowns and damage incidents
 - To evaluate driver performance
 - To study fuel efficiency and cost patterns
 - To build interactive dashboards and reports for decision-making
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3. Dataset Description

The dataset includes the following attributes:

- Shipment Details: Shipment ID, Client Name, Industry, Origin, Destination, Route

- Vehicle Details: Vehicle ID, Type (Truck/Van/Bike)
 - Driver Details: Driver Name, Experience
 - Delivery Information: Dispatch Date, Expected Date, Actual Date, Status
 - Operational Metrics: Distance, Fuel Used, Fuel Cost
 - Performance Indicators: Breakdown Flag, Damage Flag, Customer Rating
-

4. Methodology

The project was implemented using the following modules in IBM Cognos Analytics:

- Reporting
- Exploration
- Dashboard Creation
- Story Visualization

Data was analyzed using:

- Aggregation and grouping
 - Calculated metrics (SLA %, Fuel Efficiency)
 - Conditional formatting
 - Interactive filters and prompts
-

5. Implementation

5.1 Delivery Performance Report

- Grouped by Client Name
- Metrics:
 - Total Shipments
 - On-Time Deliveries
 - Delayed Deliveries
 - Failed Deliveries
 - SLA Compliance Rate (%)
- Conditional Styling:
 - < 85% → Red

- 85–95% → Amber
- 95% → Green
- Filters: Client Industry, Route Name
- Header and footer added for professional presentation

5.2 Vehicle Breakdown & Damage Report

- Fields:
 - Vehicle ID, Vehicle Type, Driver Name
 - Breakdown Count, Damage Count
 - Fuel Cost, Fuel Efficiency
 - Highlighted vehicles with Breakdown Count > 3
 - Included summary of breakdown cost impact
-

5.3 Route Performance Analysis

- Visualization: Bar/Line chart (user selectable)
 - Filters: Vehicle Type, Month
 - Highlighted routes with failure rate > 10%
 - Identified:
 - Top 3 problem routes
 - Top 3 best-performing routes
-

5.4 Exploration Analysis

- Driver Experience vs Delivery Failure Rate
- Vehicle Type vs Breakdown Frequency
- Distance vs Fuel Efficiency

Charts were annotated to highlight key patterns and insights.

5.5 Dashboard Design

The dashboard includes:

- KPI Tiles:
 - Total Shipments
 - On-Time Delivery Rate

- Breakdown Count
 - Total Fuel Cost
 - Visualizations:
 - Shipments by Client (Bar Chart)
 - Daily Delivery Trend (Line Chart)
 - Delivery Status Distribution (Pie Chart)
 - Driver Performance Table
 - Filters for Route and Vehicle Type
-

5.6 Story Creation

The story presentation consists of:

1. Operations Overview
 2. SLA Compliance Analysis
 3. Delivery Failures & Damage Analysis
 4. Breakdown Trends & Fleet Health
 5. Driver Performance & Fuel Efficiency
 6. Optimization Recommendations
-

6. Key Findings

- Experienced drivers have lower delivery failure rates
 - Trucks are more prone to breakdowns than vans and bikes
 - Longer distances tend to reduce fuel efficiency
 - Certain routes consistently show high failure rates
 - Fuel cost is a major contributor to operational expenses
-

7. Business Impact

- Improved visibility of delivery performance
- Early identification of operational risks
- Better fleet management decisions
- Enhanced SLA compliance monitoring
- Support for cost reduction strategies

8. Challenges Faced

- Understanding complex logistics dataset
 - Creating accurate calculated fields
 - Applying conditional formatting rules
 - Designing user-friendly dashboards
-

9. Conclusion

This project demonstrates how IBM Cognos Analytics can be effectively used to transform raw logistics data into meaningful insights. The developed reports, dashboards, and stories enable SwiftRoute Logistics to optimize operations, reduce costs, and improve service quality.

10. Future Enhancements

- Integration with real-time tracking systems
- Predictive analytics for breakdown prevention
- AI-based route optimization
- Advanced driver performance scoring models

Delivery Performance Report

Delivery Performance Report (Reporting)

Objective

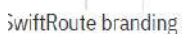
To analyze delivery performance by client and evaluate SLA compliance.

Implementation

- Grouped data by Client Name
- Calculated metrics:
 - Total Shipments
 - On-Time Deliveries
 - Delayed Deliveries
 - Failed Deliveries
 - $SLA\ Compliance\ \% = (On-Time / Total\ Shipments) \times 100$
- Applied conditional formatting:
 - $< 85\% \rightarrow$ Red
 - $85-95\% \rightarrow$ Amber
 - $95\% \rightarrow$ Green
- Added filters:
 - Client Industry
 - Route Name
- Included:
 - Header with company branding
 - Footer with "Operations Confidential"

Outcome

- Easily identified low-performing clients
- Helped monitor SLA compliance effectively



Client Name	Route Name	Client Industry	Total Shipments	On-Time Deliveries	Delayed Deliveries	Failed Deliveries	SLA Compliance%
Tata Logistics	East	Manufacturing	81	74	4	3	91.29
BlueDart	South	Retail	69	26	24	19	42.5
Delhivery	South	E-commerce	69	14	39	16	28.26
BlueDart	West	Retail	42	15	19	8	35.71
Flipkart	North	Logistics	34	34	0	0	100
Flipkart	West	E-commerce	48	24	9	15	50
Tata Logistics	West	Retail	44	14	10	20	31.47
Reliance	West	Manufacturing	36	32	3	1	88.89
BlueDart	North	Logistics	13	12	0	1	92.31
Delhivery	South	Logistics	41	19	15	7	46.34
Reliance	West	E-commerce	34	2	23	9	5.88
Delhivery	North	Manufacturing	20	0	11	9	0
Tata Logistics	North	Retail	11	4	5	2	36.36
Amazon India	West	Retail	25	21	1	3	84
Delhivery	West	E-commerce	61	35	8	18	56.08
Delhivery	East	Retail	11	1	4	6	9.09
BlueDart	South	Logistics	27	11	4	12	40.74
Delhivery	North	Manufacturing	22	19	0	3	86.36
Delhivery	East	Manufacturing	47	42	5	0	89.36
Tata Logistics	North	E-commerce	43	42	1	0	97.67

Operations Confidential

Vehicle Breakdown & Damage Report

Vehicle Breakdown & Damage Report (Reporting)

Objective

To evaluate vehicle reliability, breakdown frequency, and fuel efficiency.

Implementation

- Displayed:
 - Vehicle ID
 - Vehicle Type
 - Driver Name
 - Breakdown Count
 - Damage Count
 - Total Fuel Cost
- Created calculated field:
 - Fuel Efficiency = Distance / Fuel Used
- Applied conditional formatting:
 - Breakdown Count > 3 → Highlighted in Red
- Added summary:
 - Total breakdown cost impact in footer
- Used font:
 - Trebuchet MS, Size 11

Outcome

- Identified high-maintenance vehicles
- Provided insights into fuel usage and operational cost

Vehicle ID	Vehicle Type	Driver Name	Breakdown Flag	Damage Flag	Fuel Cost	Distance (km)	fuel efficiency	Fuel Used (liters)	Distance (km) / Fuel Used (liters)
VEH114	Bike	Driver_A	1	No	19773.31	1458	0.12921811	188.4	7.7388535
VEH499	Truck	Driver_H	1	No	8876.05	733	0.11787176	86.4	8.4837963
VEH316	Van	Driver_A	1	No	10865.41	873	0.12268041	107.1	8.1512605
VEH371	Truck	Driver_F	1	No	12198.02	1150	0.104	119.6	9.61538462
VEH292	Truck	Driver_B	1	No	8072.76	972	0.08148148	79.2	12.27272727
VEH265	Bike	Driver_B	1	No	3434.06	313	0.10958466	34.3	9.12536443
VEH326	Van	Driver_A	1	No	20103.59	1355	0.14804428	200.6	6.75473579
VEH113	Van	Driver_D	1	No	17211.44	1875	0.08821333	165.4	11.33615478
VEH104	Truck	Driver_J	1	Yes	4959.08	467	0.11798715	55.1	8.47549909
VEH130	Bike	Driver_C	1	No	16808.4	1642	0.1043849	171.4	9.57992999
VEH094	Van	Driver_G	1	Yes	2569.18	229	0.10655022	24.4	9.3852459
VEH241	Van	Driver_A	1	No	23819.47	1831	0.12359366	226.3	8.09102961
VEH454	Truck	Driver_F	1	No	24548.7	1683	0.13541295	227.9	7.3848179
VEH433	Truck	Driver_H	1	No	22362.27	1684	0.12583135	211.9	7.94714488
VEH396	Van	Driver_E	1	No	9811.58	730	0.12383562	90.4	8.07522124
VEH087	Bike	Driver_I	1	No	24657	1919	0.1238666	237.7	8.07320151
VEH177	Truck	Driver_B	1	Yes	2546.11	288	0.08923611	25.7	11.20622568
VEH418	Van	Driver_A	1	No	18448.2	1386	0.13831169	191.7	7.23004695
VEH490	Van	Driver_H	1	Yes	13875.38	1343	0.10900968	146.4	9.17349727
VEH262	Bike	Driver_E	1	No	5212.84	543	0.09871087	53.6	10.13059701
Overall - Total			200		2477298.32	209339	23.27347119	24735.4	1730.70742011

Route-wise Delivery Performance

Route-wise Delivery Performance (Reporting + Visualization)

Objective

To analyze route performance and delivery success rates.

Implementation

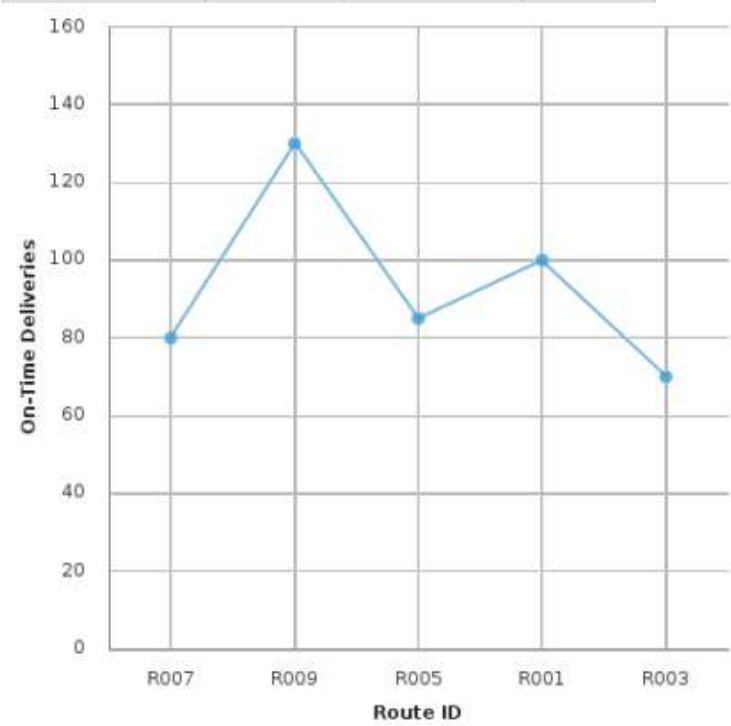
- Enabled dynamic chart selection:
 - Bar chart / Line chart
- Filters applied:
 - Vehicle Type
 - Month
- Metric used:
 - On-Time Delivery Rate
- Conditional formatting:
 - Failure Rate > 10% → Red
- Added annotations:
 - Top 3 problem routes
 - Top 3 best-performing routes

Outcome

- Highlighted inefficient routes
- Supported route optimization decisions

The screenshot shows the ITM3 software interface. At the top, there is a toolbar with various icons. Below the toolbar, a window titled 'Deselect all' is open, displaying a list of vehicle types: Jan, Truck, Van, and Car. The 'Jan' option is currently selected, indicated by a radio button. The 'Truck' option is also visible, with a radio button next to it. The 'Van' and 'Car' options are also listed. At the bottom of the window, there is a 'Deselect all' button. The main area of the software is currently empty.

On-Time Deliveries	on time rate	Failed Deliveries	FailureRate
902	9.02	153	1.53



Route ID	On-Time Deliveries
R009	130
R006	120
R001	100
R002	92
R010	90
R005	85
R007	80
R004	75
R003	70
R008	60

Data Exploration Analysis

Data Exploration Analysis (Exploration)

Objective

To discover relationships between key operational factors.

Analysis Performed

- Driver Experience vs Delivery Failure Rate
- Vehicle Type vs Breakdown Frequency
- Distance vs Fuel Efficiency

Findings

- More experienced drivers → Lower failure rates
- Trucks → Higher breakdown probability
- Longer distances → Lower fuel efficiency

Outcome

- Provided data-driven insights for fleet and driver management

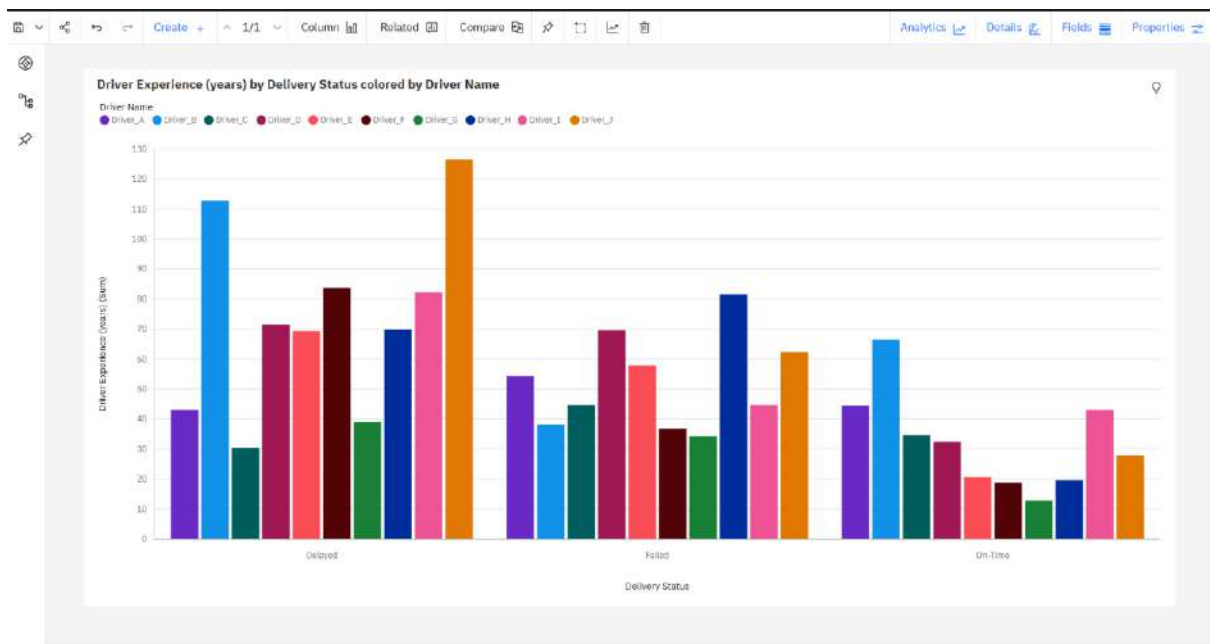


Chart A

Damage Flag by Driver Experience (years) colored by Driver Name

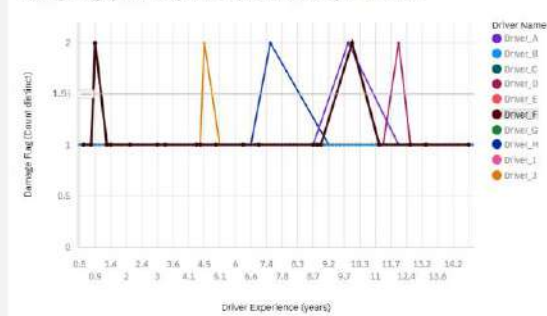
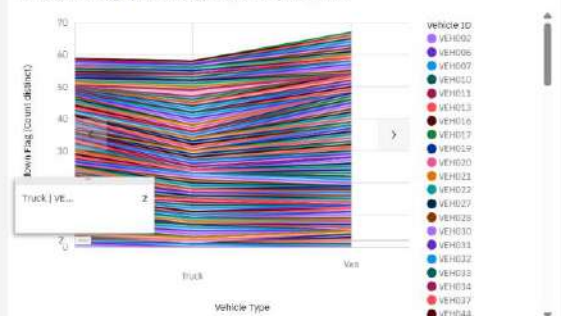
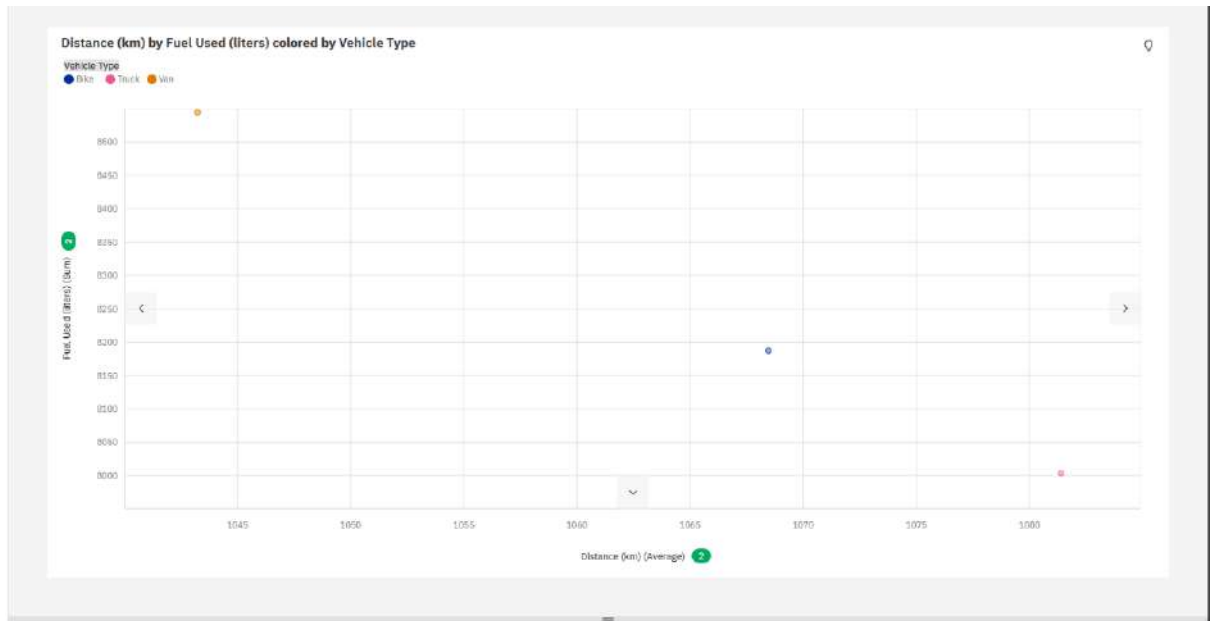


Chart B

Breakdown Flag by Vehicle Type colored by Vehicle ID



Summary	Chart A : Damage Flag	Chart B : Breakdown Flag	Combined
Minimum	1	1	-
Maximum	2	2	-
Chart percent of data set	-	-	-
Chart total	(no value)	(no value)	-
Difference of chart totals			-



Fleet Operations Dashboard

Fleet Operations Dashboard (Dashboard)

Objective

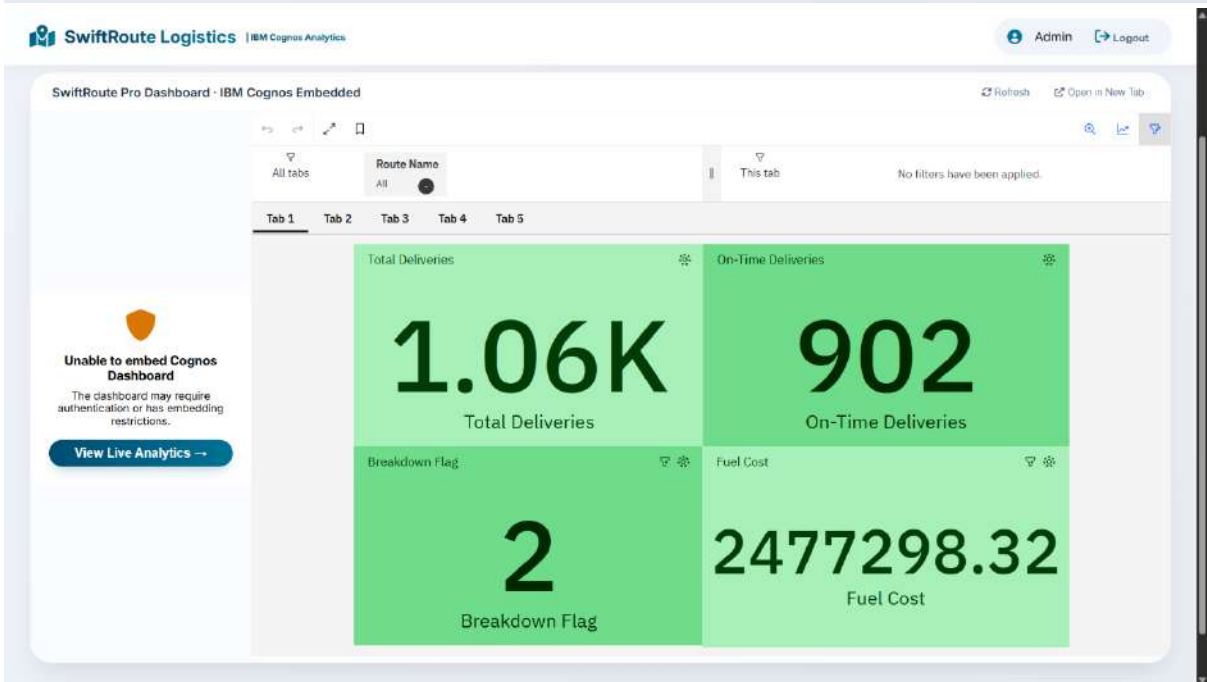
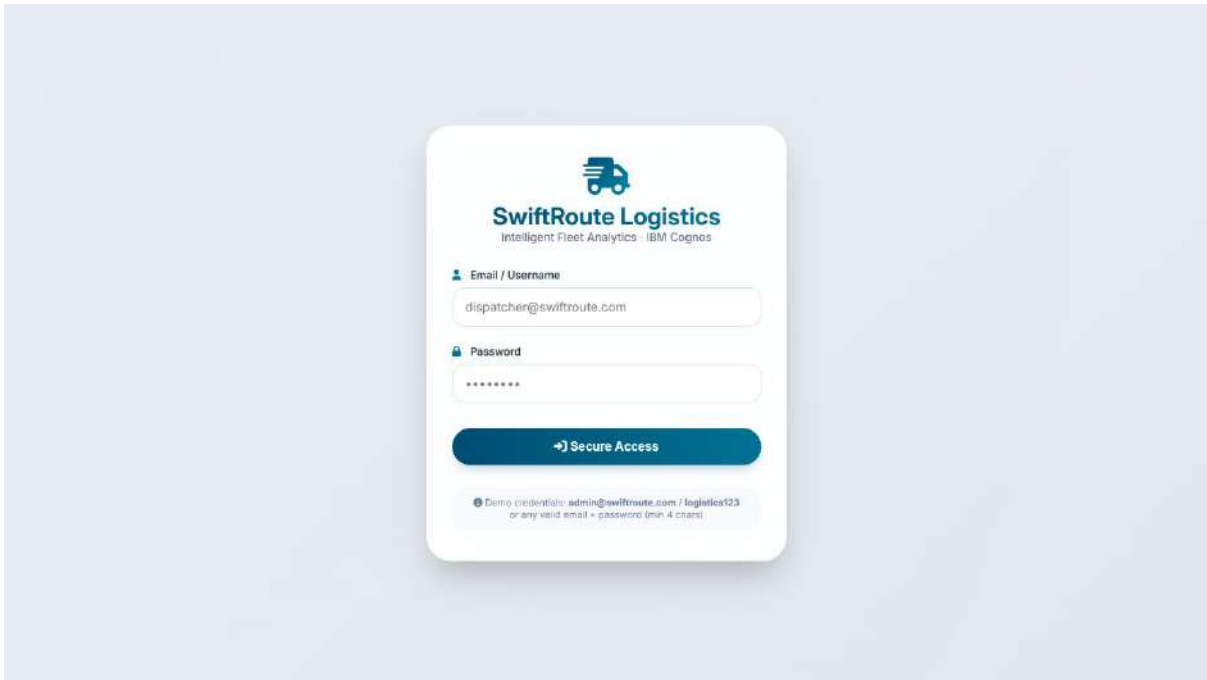
To create an interactive dashboard for real-time monitoring.

Components

- KPI Tiles:
 - Total Shipments
 - On-Time Rate %
 - Breakdown Count
 - Total Fuel Cost
- Charts:
 - Bar chart → Shipments by Client
 - Line chart → Daily Delivery Trend
 - Pie chart → Delivery Status Distribution
- Table:
 - Driver Performance by Rating
- Filters:
 - Route
 - Vehicle Type

Outcome

- Delivered centralized operational visibility
- Enabled quick decision-making



SwiftRoute Pro Dashboard - IBM Cognos Embedded

Refresh Open in New Tab

All tabs

Route Name
All

This tab

No filters have been applied.

Tab 1

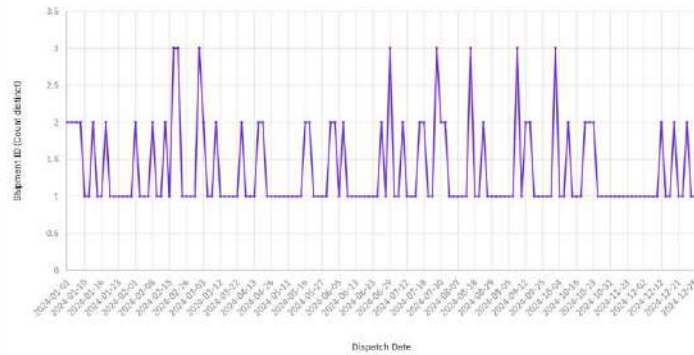
Tab 2

Tab 3

Tab 4

Tab 5

Shipment ID by Dispatch Date



Unable to embed Cognos Dashboard

The dashboard may require authentication or has embedding restrictions.

[View Live Analytics →](#)

SwiftRoute Pro Dashboard - IBM Cognos Embedded

Refresh Open in New Tab

All tabs

Route Name
All

This tab

No filters have been applied.

Tab 1

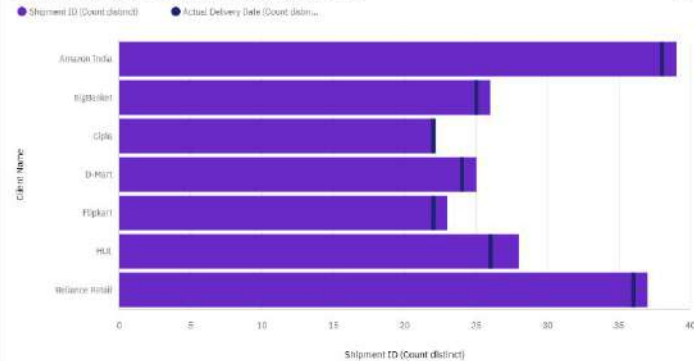
Tab 2

Tab 3

Tab 4

Tab 5

Shipment ID compared to Actual Delivery Date by Client Name



Unable to embed Cognos Dashboard

The dashboard may require authentication or has embedding restrictions.

[View Live Analytics →](#)

SwiftRoute Pro Dashboard - IBM Cognos Embedded

Refresh Open in New Tab

All tabs

Route Name
All

This tab

Vehicle Type
All

Tab 1 Tab 2 Tab 3 Tab 4 Tab 5

Driver Name, Customer Rating (1-5), Shipment ID, Delivery Status and Damage Flag

Driver Name	Customer Rating (1-5)	Total Deliveries	On-Time Deliveries	Failure Rate
Driver_A	2.787	8	Delayed	No
	1.1	1	Delayed	Yes
	2.6	9		
	2.567	6	Failed	No
	2.4	1	Failed	Yes
	2.543	7		
	2.88	5	On-Time	No
	2.88	5		
	2.767	9	Delayed	No
	2.45	4	Delayed	Yes
Summary	2.886	21		
Driver_B	2.767	9	Delayed	No
	2.45	4	Delayed	Yes
	2.669	13		
Driver_C	3.05	2	Failed	No
	2.4	1	Failed	Yes

Unable to embed Cognos Dashboard

The dashboard may require authentication or has embedding restrictions.

[View Live Analytics →](#)

SwiftRoute Pro Dashboard - IBM Cognos Embedded

Refresh Open in New Tab

All tabs

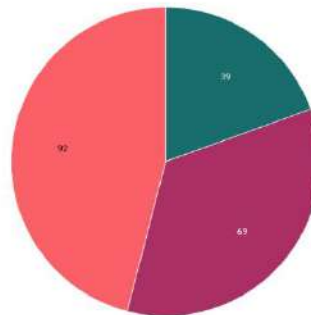
Route Name
All

This tab

No filters have been applied.

Tab 1 Tab 2 Tab 3 Tab 4 Tab 5

Shipment ID by Delivery Status

Delivery Status
On-Time Failed Delayed

Unable to embed Cognos Dashboard

The dashboard may require authentication or has embedding restrictions.

[View Live Analytics →](#)

Monthly Fleet Review Story

Monthly Fleet Review Story (Story Making)

Objective

To present insights in a storytelling format for management.

Scenes

1. Operations Overview & Fleet Statistics
2. Client SLA Compliance Performance
3. Delivery Failure & Damage Analysis
4. Vehicle Breakdown Trends & Fleet Health
5. Driver Performance & Fuel Efficiency
6. Route Optimization & Cost Reduction

Outcome

- Improved communication of insights
- Supported strategic planning and decision-making

Overall Project Conclusion

The SwiftRoute Logistics project successfully demonstrates the practical application of IBM Cognos Analytics in solving real-world logistics and supply chain challenges. By integrating reporting, exploration, dashboard, and storytelling modules, the project provides a comprehensive analytical solution that transforms raw operational data into meaningful business insights.

The analysis clearly shows that logistics operations are highly dependent on multiple interconnected factors such as driver experience, vehicle condition, route efficiency, and fuel consumption. Through systematic data visualization and analysis, the project uncovers inefficiencies and highlights areas that require immediate attention.

Overall, this project proves that data-driven decision-making can significantly enhance operational efficiency, reduce costs, and improve service quality in the logistics industry.

Final Business Impact Conclusion

The overall analysis indicates that SwiftRoute Logistics can achieve significant improvements by adopting data-driven strategies. Key benefits include:

-  Reduction in delivery failures and delays
-  Improved fleet utilization and maintenance planning
-  Better fuel cost management
-  Enhanced driver performance and allocation
-  Increased SLA compliance and customer satisfaction

By leveraging IBM Cognos Analytics, the organization can move from reactive problem-solving to proactive decision-making.

Final Statement

In conclusion, this project highlights the importance of business intelligence in modern logistics operations. The integration of analytics into daily operations enables SwiftRoute Logistics to optimize performance, reduce costs, and maintain high service standards. The project not only fulfills its analytical objectives but also demonstrates strong capabilities in data visualization, interpretation, and business problem-solving.